



Clinical Case Report

Fatal delayed cardiac tamponade due to rupture of micropseudoaneurysm of left anterior descending coronary artery following stab to the chest



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ABSTRACT

Traumatic coronary pseudoaneurysm has been described to be mainly associated to iatrogenic lesion of the coronary arteries. However, chest-stab-wound-related coronary pseudoaneurysm caused by isolated partial incision of a coronary artery giving rise to fatal delayed cardiac tamponade is very rare. We describe an autopsy case in which this potentially fatal complication developed 8 days later after a thoracic stab wound. Unfortunately, the imaging examination failed to detect this defect during hospitalization. Postmortem examination revealed that the posterior wall of the left anterior descending coronary artery was intact but that the anterior wall was incised, forming a micropseudoaneurysm which had ruptured. This case highlights that isolated coronary artery injuries must be considered in any patient with a penetrating wound to the thorax, and coronary pseudoaneurysms should not be missed in these patients.

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1. Introduction

Coronary artery aneurysm is rare in adult patients, and atherosclerosis is the most common cause followed by inflammation, infections, and connective tissue disorders [1]. It is classified as true or false (pseudo) based on the respective presence or absence of all three layers (intima, media, and adventitia) in the wall of the aneurysm [2]. Coronary pseudoaneurysm is much less common than true aneurysm and usually occurs after catheter-based coronary intervention as a result of traumatic dissection or perforation of the coronary artery [1,3]. Coronary artery pseudoaneurysm can also occur spontaneously, due to blunt chest trauma, and in association with cardiac tumors [1,3]. However, isolated penetrating injury of a coronary artery causing coronary pseudoaneurysm and fatal delayed cardiac tamponade is very rare.

We report a fatal delayed cardiac tamponade due to rupture of a micropseudoaneurysm of the left anterior descending (LAD) coronary artery caused by isolated partial abscission following thoracic stab.

2. Case report

2.1. Clinical history

An 18-year-old male was stabbed in his left thorax by a dagger. On admission, an emergency computed tomography (CT) scan showed left pleural effusion and small rupture in the inferior lobe of left lung.

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Echocardiography, dynamic electrocardiogram, chest X-ray, electrolytes, serum cardiac enzymes, and blood coagulation assays all showed no abnormalities. On the third day of admission, a repeated CT scan and echocardiography both revealed similar findings to those obtained previously. But coronary angiogram was not performed. He markedly improved with effective antibiotic treatment and thoracic drainage during hospitalization. However, 8 days later, he suddenly collapsed and went into cardiopulmonary failure and died after in about 30 min following the collapse.

2.2. Autopsy findings and histopathology examination

An autopsy was performed 28 h after death. A 2.2-cm-wide stab wound was found 13.5 cm from the median sternal line at the fourth intercostal space. The anterior surface of the pericardium had no stab wound but a 2-cm-long incision on the inner wall. The pericardial cavity was filled with 570 ml of clotted blood. The middle segment of the LAD coronary artery on the anterior interventricular surface had a rounded excrescence (4 mm in diameter) with a superficial cross stab and a tiny break (Fig. 1a). The transverse section of the excrescence showed an aneurysmal dilatation of the LAD coronary artery with an occlusive hematoma (Fig. 1b). On histology examination, serial sections (Fig. 2) revealed that the posterior coronary wall was intact but that the anterior wall had ruptured, extending to the adjacent hematoma (organized thrombus) and forming an hourglass-shaped aneurysmal defect. Loss of vessel wall integrity with organized intramural thrombus formation was present but without the internal elastic lamina or smooth muscle. The true and false lumens were communicated by the ruptured wall. Besides, the ruptured wall presented hyperplasia of some collagen fibers. No penetrating lesions and myocardial infarction of the ventricular

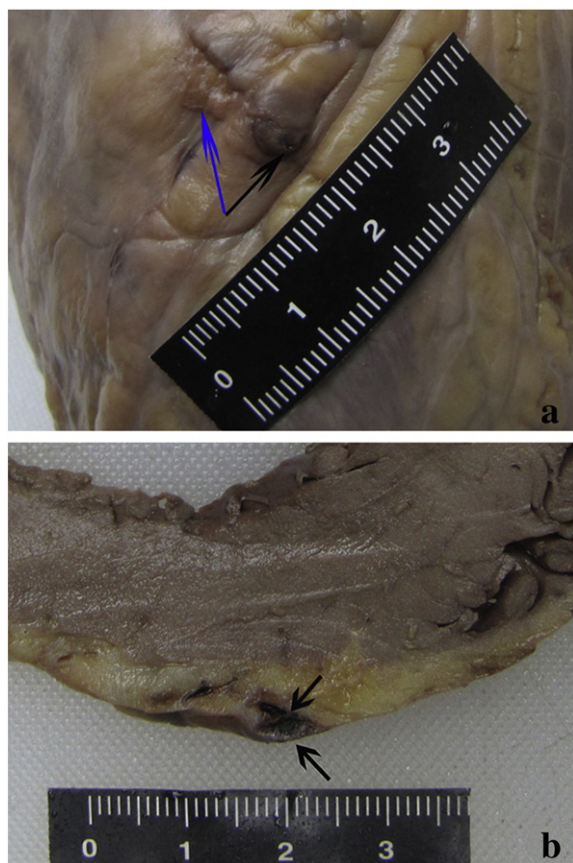


Fig. 1. (a) The epicardial position of the LAD coronary artery showing the excrescence with a superficial cross stab (blue arrow) and a tiny break (black arrow). (b) Pseudoaneurysm formation of the LAD coronary artery and its external rupture (lower arrow). Note the thin-walled vascular structure (upper arrow) between the coronary artery and the pericoronary hematoma.

walls were detected. The other coronary arteries, cardiac valves, and cardiac conduction system were all normal. There was a 3.0×2.0-cm puncture with hemorrhage in the inferior lobe of left lung.

3. Discussion

Coronary pseudoaneurysm is defined as a disruption of coronary vessel wall integrity and the formation of a closed space surrounded by adjacent structures [4]. Unlike the true aneurysm with the presence of a wall containing intima, media, and adventitia, a pseudoaneurysm shows a loss of vessel wall integrity and transition from a three-layered wall to an outwardly bulging monolayer [3]. According to the pathological features in this case, a traumatic coronary pseudoaneurysm was diagnosed.

Blunt chest-trauma-related coronary pseudoaneurysms and ventricular pseudoaneurysms due to penetrating chest wound have been reported [1,5–8]. However, chest-stab-wound-related coronary pseudoaneurysms are rare, and only one case has been reported [4]. Since selective penetrating injuries to the coronary arteries are rare, accounting for less than 5% of all the penetrating cardiac injuries [9,10], coronary pseudoaneurysms caused by such injuries are even rarer. In addition, the occurrence of a delayed tamponade after coronary vessel injury is rare and usually coupled with many complications, including myocardial infarction, valvular injury, coronary–cameral fistula, and many others [11–14].

In this case, the LAD coronary artery was not ruptured completely; only the anterior wall was involved following the stab wound. The spasm and contraction of the injured artery due to nerve reflex at the time of injury or extravasation of arterial blood from the injured artery

being surrounded by the adjacent epicardium were likely to seal the incision, which might lead to the formation of a coronary micropseudoaneurysm with intramural thrombus to occlude the cardiac wound and stem the bleeding. However, the pseudoaneurysm ruptured, resulting in a fatal cardiac tamponade. The presence of organized aneurysm wall and hyperplasia of collagen fibers in the ruptured vascular wall indicated that the injury and the formation of the pseudoaneurysm had underwent a period of time, consistent with the clinical history.

This case had several unique clinical features. The patient showed stable hemodynamics and normal electrocardiogram and echocardiography during hospitalization. Unlike previously reported cases [12–14], the normal levels of serum cardiac enzymes might be due to the isolated injury of the LAD coronary artery without myocardial damage. Especially, the most notable characteristic of this case is that the primary damage to the LAD coronary artery was minor, resulting in a micropseudoaneurysm which ruptured suddenly followed by delayed cardiac tamponade. The delayed cardiac tamponade was attributed to the temporary hemostasis exerted by the micropseudoaneurysm. Since imaging evidence of pericardial hemorrhagic effusion and ischemic electrocardiographic changes were absent, the diagnosis was only made by postmortem examination.

Though progress in the management of penetrating cardiac injury has saved many patients [15], penetrating cardiac injury remains a highly lethal form of trauma today [9]. The therapeutic approach consists of vessel repair, specifically, artery ligation or aortocoronary bypass [16]. Echocardiography is considered to be the first diagnostic approach on initial assessment and follow-up, especially in hemodynamically stable patients [13], but it is not infallible [9], as in this case. Thus, the absence of pericardial effusion does not rule out cardiac injuries especially coronary artery injuries. Since delayed or missed diagnosis and treatment are lethal, high-quality imaging techniques and expertise are vital for the prompt diagnosis and further management. This case emphasizes that coronary artery injuries must be kept in mind in all patients with penetrating thoracic trauma. In addition, coronary pseudoaneurysms and delayed cardiac tamponade must be vigilantly monitored as potential complications in these victims.

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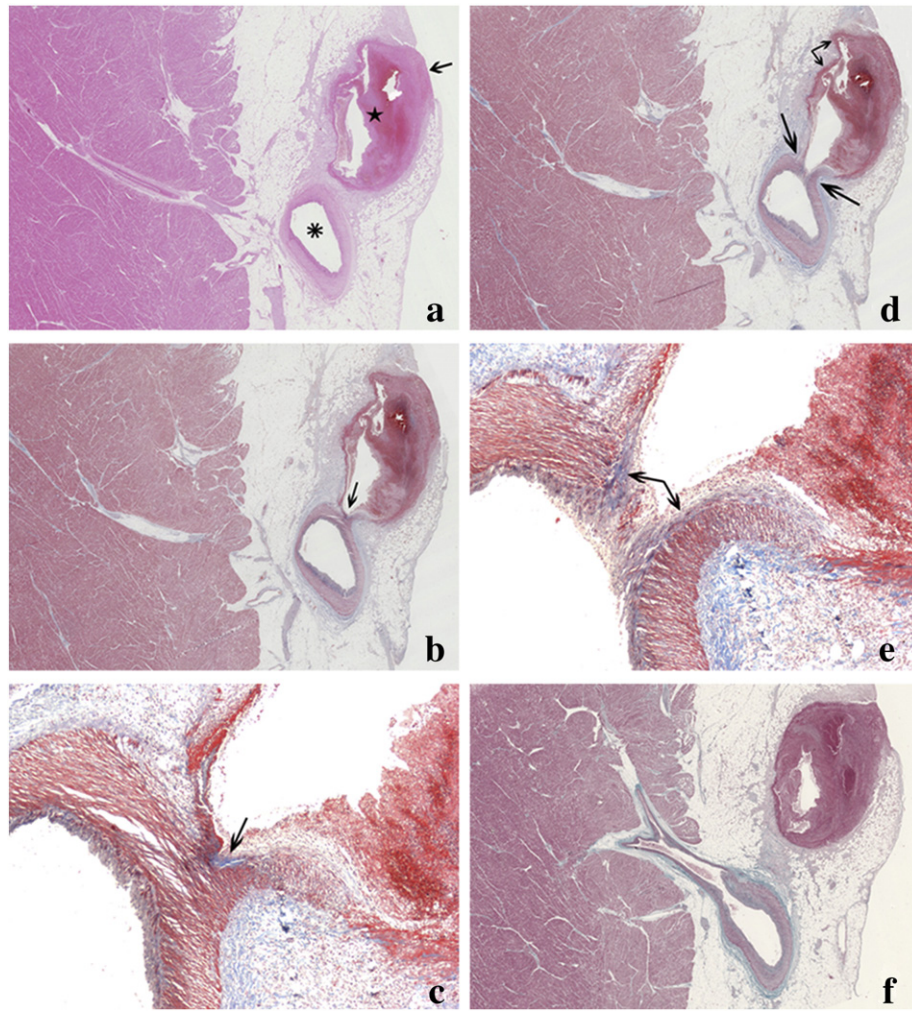


Fig. 2. Histological examination of the coronary lesion on serial sections. (a) An occlusive thrombus (pentagram) with organized adventitia beside the LAD coronary artery (asterisk). The ruptured epicardial surface was seen (arrow). (b) The adventitia and partial media of the anterior wall of the LAD coronary artery were incised (arrow). (c) Higher amplification of the partially incised artery showing hyperplasia of collagen fibers (arrow). (d) Completely ruptured vascular wall (large arrows) extended to the hematoma forming the neck of the pseudoaneurysm. The thin-walled pseudoaneurysm wall had no internal elastic lamina or smooth muscle (small arrows). (e) Higher amplification of the completely ruptured anterior wall of the artery (arrows). (f) Further histology showing the occlusive thrombus isolated from the artery again [(a) hematoxylin and eosin, $\times 10$; (b–f) Masson's trichrome, (b, d, and f) $\times 10$, (c and e) $\times 40$].

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